KARNAJEET GOSAVI

+91-9320754541 | karnajeet.gosavi23@vit.edu | linkedin.com/in/karnajeet-gosavi | github.com/kg290 | portfolio-kg290s-projects.vercel.app

SUMMARY

Python-focused AI engineering candidate building systems that turn complex documents, security signals, and long-form knowledge sources into reliable automated workflows. Hands-on across agentic reasoning, retrieval pipelines, LLM verification, FastAPI services, PDF parsing, adaptive routing, Docker-backed deployment, and evaluation-driven backend design.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Observability Professional; Oracle Cloud Infrastructure 2025 Certified Generative AI Professional; IBM DevOps.

EDUCATION

Vishwakarma Institute of Technology, Pune - B.Tech in Computer Science Engineering | Jul 2023 - Jun 2027
GPA: 8.98/10

SKILLS

Programming: Java, Python, JavaScript, TypeScript, C++, C#, PowerShell 7
Backend: Django, Flask, FastAPI, PostgreSQL, MySQL, MongoDB, Redis, ASP.NET Core, Spring Boot, REST & GraphQL API Design
Frontend: React.js, Next.js, Flutter, Tailwind CSS, Framer Motion, Three.js, D3.js
AI/ML: PyTorch, TensorFlow, LangChain, FAISS, OpenCV, Deep Learning
DevOps/Cloud: Git, Docker, AWS, GCP, Azure, GitLab
OS: Windows Server, Kali Linux, Ubuntu

EXPERIENCE

AI Engineer - Industry Project | Syngenta | Jan 2026 - Apr 2026
Python, SAP CPI, BPMN/XML, Generative AI, DOCX Automation

- Engineered a SAP CPI specification generator that converts 7 artifact groups - iFlow packages, BPMN/XML flows, XSD/WSDL schemas, mappings, adapters, Groovy scripts, and external parameters - into one structured DOCX output.
- Classified 5 integration layers - main flows, local subprocesses, message routes, adapter metadata, and error-handling paths - before generation to improve traceability across SAP project files.
- Generated 10+ documentation sections with 4 diagram views, artifact inventories, schema summaries, code blocks, headers/footers, security details, mappings, and exception-flow coverage.
- Automated a 4-step documentation workflow - artifact discovery, parsing, AI summarization, and DOCX formatting - while summarizing up to 80 text artifacts per package for dependency and risk review.

PROJECTS

Narrative Consistency Verification System | Python, Agentic AI, LLM Reasoning, Hybrid Retrieval, FAISS, Pathway

- Engineered a multi-stage narrative verification pipeline that converts character backstories into typed atomic claims across timeline, family, location, relationship, event, occupation, and historical-reference categories.
- Designed hybrid evidence retrieval using semantic similarity, keyword overlap, character profiling, and ranked passage selection to ground LLM reasoning against long-form source novels exceeding 100k words.
- Architected a layered verifier combining deterministic contradiction rules, historical consistency checks, claim-level LLM reasoning, confidence aggregation, and early termination for high-confidence contradictions.
- Delivered batch verification workflows with structured outputs, evidence traces, confidence scores, contradiction labels, and CSV reporting to support repeatable evaluation.

Honey Proxy Deception Firewall | Python, Flask, React, TypeScript, SQLAlchemy, Docker

- Engineered an adaptive deception firewall that forwards live traffic through a Flask proxy, maintains per-session state, and dynamically routes suspicious or hostile users into monitored deception flows.
- Engineered a weighted behavioral scoring engine across 16 signals, including burst velocity, endpoint probing, payload entropy, failed-login ratio, automation timing, hidden-admin access, source reputation, and sequential route discovery.
- Designed rolling baselines and session windows to compare current behavior against historical request patterns, reducing reliance on static rule matching and allowing score changes across a session.
- Modeled SQLAlchemy event tables, honeypot ingestion APIs, incident report generation, analytics snapshots, and a React dashboard for live traffic monitoring, session drilldowns, and attack investigation.

Term Sheet Validator | Python, FastAPI, PyMuPDF, AsyncIO, Pydantic, LLM APIs

- Architected a FastAPI-based document intelligence service that ingests PDF term sheets, extracts text with PyMuPDF, chunks clauses, and answers structured finance/legal queries through an API endpoint.
- Designed clause-aware chunking with section detection, priority scoring, overlap windows, deduplication, and query-specific retrieval to preserve context across dense legal and financial documents.
- Designed an adaptive LLM routing layer with a 6-key API pool, cooldown handling, token estimation, fallback model selection, and complexity-based request timing for concurrent question answering.
- Integrated confidence-aware response formatting, answer cleanup, JSON output validation, and async processing to make the system behave like a reliable backend service instead of a single prompt wrapper.

NOTABLE ACHIEVEMENTS

- Published 3 research papers in IEEE and Springer venues.
- Technical and Associate Secretary, MLSC and Student Development Club 2025; coordinated technical events, workshops, and student development initiatives.
- Won 2+ national-level hackathons across AI, software engineering, and innovation competitions.